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Temperature Effect on Load Distribution, Friction, and Wear of a Grease-Lubricated Spherical Roller Bearing (SRB)

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ABSTRACT

This article investigates the effect of temperature on the lubrication and wear behavior of a grease-lubricated spherical roller bearing (SRB) assuming mixed-elastohydrodynamic lubrication (mixed-EHL) where the contact load is shared between the lubricant film and contact asperities. Temperature effects on internal load distribution and elastic modulus and their subsequent effect on the wear are also analyzed. The effect of temperature is investigated, first on the film thickness and load sharing, and then on the friction and wear. If the temperature rises from ambient to 110°C, the increase in the contact patch dimensions, due to the decrease of elastic modulus, is only approximately 1% and the decrease in mean contact pressure is approximately 2%. Moreover, the load at $\psi = 0^\circ$ decreases by approximately 4%, and the load zone increases by approximately 5%. As the temperature rises from 50 to 110°C, the asperity portion of the contact load at $\psi = 0^\circ$ position is increased by 350%. At 50°C, only 20% of the total traction is contributed by the contact asperity, which is 80% at 110°C. For the temperature rise from 50 to 110°C, the wear volume loss increases by 238% and nonlinearly, due to viscosity reduction and its effect on the reduction of film thickness. The isolated change of internal load distribution and elastic modulus due to temperature rise causes a relatively insignificant change in wear. Hence, the predominant effect of temperature rise on wear is due to a decrease in lubricant film thickness causing the asperity component of the contact load to increase.

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Friction coefficient; wear; rolling element bearing; load sharing; mixed lubrication; temperature–internal load distribution; temperature–elastic modulus

Introduction

Spherical roller bearings are critical components of many machines like rail vehicles, wind turbine gearboxes, paper mills, and so on. The failure of these bearings significantly harms the machine's performance and life (1–5). Besides surface and subsurface initiated fatigue (e.g., micropitting or spalling) (6–8), wear is a major cause of rolling bearing maintenance. The lubricants are usually considered an integral part of any rolling bearing and provide a film between contacting surfaces to prevent metal-to-metal interaction. The breakdown of the film increases the metal-to-metal contact, which increases the wear of a contact. The bearing operating temperature significantly influences the lubricant film buildup and hence the separation of the mating surfaces, and thus wear strongly depends on temperature. Moreover, the internal load distribution is a fundamental feature of all rolling element bearings, and the determination of it is the first step in modeling wear. Other than the magnitude of externally applied load and number of rolling elements, the bearing internal load distribution largely depends on the operating internal clearance (9), that is, a free space provided between rolling elements and raceways, which also plays a role in the movement of rolling elements. Increasing temperature can cause an expansion of the bearing

structures, which can further alter the internal clearance and finally the internal load distribution (10, 11). The internal clearance is changed based on the magnitude of temperature and on bearing/housing/shaft materials (10–13). Radial internal clearance strongly affects noise, vibrations, fatigue (14, 15), and possibly wear, as its change can cause a change in internal load distribution. Hence, diametral clearance needs to be calculated at the operating temperature to predict the internal load distribution accurately. Furthermore, temperature also influences the material properties, particularly the elastic modulus, an important parameter for contact analyses (16). The elastic modulus decreases if the body temperature rises, and this can influence the traction force and contact patch of a rolling contact problem (17). Hence, the change of elastic modulus induced by temperature should be considered in the analysis of contact problems. Therefore, bearing wear investigation requires the consideration of the change in internal load distribution and elastic modulus with temperature, although previously it was uncertain how much effect this can have on wear.

Rolling bearings are mostly lubricated with grease, and failure of the grease is considered the predominant cause of bearing failure (18, 19). Grease consists of base oil, additives, and thickener microstructure and acts as an oil reservoir that releases oil slowly to replenish the contacts (20, 21).

Nomenclature

A_H	Hydrodynamic area of the contact	R'	Reduced radius of the contact m
A_{nom}	Nominal contact area	s_0	Temperature–viscosity coefficient 1.246
a and b	Half contact length in the rolling direction and transverse direction respectively m	T_a, T_b, T_c , and ΔT	Ambient temperature, bulk temperature, contact temperature, and flash temperature rise, respectively (°C)
b_{fiber}	Fiber diameter 100×10^{-9} m	t	Constant used in load distribution equation 2/3
C_p	Specific heat of contacting bodies 460 J/kg°C	$u_{r,0}$	Initial radial clearance 1.9×10^{-4} m
d_d	Summit-mean surface plane separation ($d_{d,r}$: 41×10^{-9} $d_{d,i/o}$: 34×10^{-9}) m	$u_{r,w}$	Working radial clearance m
$E(20)$	Elastic modulus at 20 °C 207×10^9 Pa	u_s	Sliding speed of the contact m/s
$E(T_b)$	Elastic modulus at T_b Pa	$W_{1,A_{i/o}}$	Average local wear profile change
E'	Equivalent elastic modulus 227×10^9 Pa	w	Roller width 27.6×10^{-3} m
F_A and F_H	Asperity and hydrodynamic portion of the contact load N	Z_r	Number of rollers in each side 22
F_R	Radial load 10750 N	α	Pressure–viscosity coefficient 27.5×10^{-9} Pa ⁻¹
F_T	Total contact load N	α_r, α_i , and α_o	Thermal expansion coefficients of roller, inner race, and outer race 12.5 K ⁻¹
f_{max}	Maximum fiber volume fraction $3\pi/16$	β	Mean asperity summit radius β_r : 15×10^{-6} $\beta_{i/o}$: 21×10^{-6} m
f_{mo}	Initial fiber mass fraction 0.29	γ_A and γ_H	Asperity load factor and hydrodynamic load factor
f_x	Traction in the rolling direction	$\dot{\gamma}$	Shear strain rate
G	Shear modulus 79×10^9 Pa	ε_T	Total radial clearance change m
h	Film thickness	$\varepsilon_{T,r}$ $\varepsilon_{T,i}$, and $\varepsilon_{T,o}$	Thermal expansion of roller, inner raceway, and outer raceway
k	Thermal conductivity of bearing steel 47 watts/mK	η	Base oil viscosity Pa.s
K	Frictional power wear coefficient Kg/(N.mm)	η_0	Ambient base oil viscosity (40 °C, 101.3 kPa) 0.0837 Pa.s
L_t	Surface length of initial grease profile 0.1481 m	θ_r	Roller angle 11.5 °
l_t	Track length 2.525 m	μ	Friction coefficient
m_r	Mass of rollers 65.6×10^{-3} kg	μ_A and μ_H	Asperity and hydrodynamic component of friction coefficient
n_a	Density of asperity $n_{a,r}$: 6.7×10^7 $n_{a,i/o}$: 5.66×10^{10} m ⁻²	μ_b	Boundary friction coefficient —
p_A and p_H	Asperity and hydrodynamic part of the contact pressure respectively Pa	ν	Poisson's ratio 0.3 —
p_h and p_{mean}	Maximum and mean Hertzian contact pressure Pa	ξ	Creep ratio
Pe	Peclet number	ξ_{avg}	Mean of absolute creep ratio across the contact
p_T	Total contact pressure Pa	ρ_0	Ambient lubricant density Grease: 910, oil: 900 kg/m ³
Q_ψ	Roller load at an angle ψ N	ρ_s	Density of bearing material 7800 kg/m ³
Q_0	Roller load of maximally loaded roller position ($\psi = 0^\circ$) N	σ_s	Standard deviation of asperity heights $\sigma_{s,r}$: 67×10^{-9} $\sigma_{s,i/o}$: 41×10^{-9} m
q	Average heat flux of the contact watts/m ²	τ_H	Hydrodynamic shear stress
q_A and q_H	Asperity and hydrodynamic component of heat flux watts/m ²	τ_0	Eyring shear stress
$R_{x,r}$	Radius of roller in the rolling direction	ψ	Roller azimuth position (defined as $-\pi \leq \psi \leq \pi$)
$R_{x,i}$	Inner race radius in the rolling direction	χ and ζ	Elliptic integral of first and second kind
$R_{x,o}$	Outer race radius in the rolling direction	ω_r	Angular velocity of rollers
$R_{y,r}$	Radius of roller in the transverse direction		
$R_{y,i}$	Inner race radius in the transverse direction		
$R_{y,o}$	Outer race radius in the transverse direction		

The process of oil release from the grease microstructure is often called "grease bleeding." In addition, the preference of grease for rolling bearings is due to its oil bleeding ability and preventive behavior in terms of leaking out (18). An oil bleeding model for grease lubricants based on Darcy's law of viscous flow was developed by Baart et al. (22). Here, the bleeding rate depends on the pressure gradient, oil viscosity, and permeability of the thickener network (21, 23). In a loaded contact, the grease may be lost from the track due to high contact pressure. Zoelen et al. (24) have developed a model for the pressure-induced side flow loss of a grease-lubricated rolling element bearing assuming a starved

condition. Later, Pozzebon et al. (25) integrated the Zoelen et al. (24) pressure-induced side flow loss model with the Baart et al. (22) oil bleed model to calculate the film thickness in a rolling element bearing to predict the wear of a spherical roller bearing by accounting for the grease effect. Grease performance improvement is the best approach to reduce bearing wear, and many researchers are investigating using different types of additives. For example, an improved wear performance of bearing material was obtained when tested using grease with nanoparticles (26). Nehme (27) studied the wear of a bearing using grease with/without MoS₂ for different load and concentration. The results

showed that for an extreme pressure application, the same improved wear performance can be found with a reduced concentration of MoS₂. Furthermore, improved wear performance of oil lubricant was obtained using other different types of additives (28–30) that can also be investigated for the case of grease.

Rolling element bearings are mostly run under mixed-elastohydrodynamic lubrication (mixed-EHL) conditions where surface roughness and lubricant play a significant role in the tribological performance (31). The major characteristics of a mixed-EHL contact are where (i) asperity and hydrodynamic contact coexist, (ii) wear is mainly contributed by the asperity component of the contact load, and (iii) the friction coefficient is also composed of asperity and hydrodynamic components (32–37). The temperature influences load sharing between lubricant and surface roughness through the lubricant's rheological properties. Much research has been performed on the wear modeling of mixed lubricated contact (37–40), including one study for a grease-lubricated rolling element bearing by Pozzebon et al. (25). The Pozzebon et al. model successfully predicted the wear of a grease-lubricated bearing; however, the model did not consider the role of the thermal effect on the internal load distribution or any other localized contact parameters. Zhou et al. (38) developed a wear model for gear application by considering the contact surface temperature and showed lower wear at lower contact temperature. The Pei et al. (39) wear model for a sliding contact showed that the wear volume increases with temperature. Zhao et al. (40) developed a wear model for Cu-based friction pairs assuming mixed-lubricated contact in which wear loss, actual contact area ratio, and the load-sharing ratio were investigated for different velocities, temperatures, and pressure. Among all these parameters, the effect of temperature on wear was the most significant. As shown by Zhao et al. (40), for a temperature rise from 30 to 150 °C, wear loss was increased from less than 0.4 to 2.3 mg. The Rosenkranz et al. (41) investigation on the wear preventive capabilities of different greases in rolling bearings showed a strong relation between wear performance and temperature. The lubricant molecules lose the ability to be attached to the surfaces, aggravating wear if the contact temperature is large enough (38, 42). As already discussed, the temperature can influence many parameters of a lubricated bearing, such as the rheological and tribochemical properties of grease, and the material as well as the structural properties of the bearing. Viscosity is the most significant property of a lubricant, and an increase in temperature causes a decrease in viscosity that can reduce the film thickness, further increasing the wear of a bearing. However, to the best knowledge of these authors, there is no published paper that has quantified the level of temperature related wear of a spherical roller bearing, due to its effect on grease rheological properties.

This article presents an investigation of the effect of temperature on the wear of the grease-lubricated spherical roller bearing under the mixed-lubricated condition. The model calculates the film thickness by taking into account a mass-balance approach between lubricant side flow loss and

centrifugally driven bleed supply as per Pozzebon et al. (25), in which the grease rheological properties are the input parameters. Hence, the article mainly presents wear under temperature due to its effect on grease rheology. In the model, the effects of temperature on the internal load distribution and elastic modulus are also considered. Moreover, the lubricant-asperity load sharing and friction/traction of the bearing contact were also analyzed by considering the effect of operating temperature.

Modeling

This section presents the modelling of the effect of temperature on the wear of a grease-lubricated spherical roller bearing. Figure 1 shows a schematic of a spherical roller bearing, including its components (an inner race, an outer race, two cages, and two rows of rollers). The basis of this work is the Pozzebon et al. (25) wear model for the grease-lubricated spherical roller bearing, so this section provides an overview of it.

In this study, the film thickness is calculated assuming a starved contact condition by considering the combined effect of pressure-induced side flow loss and centrifugally induced bleed supply, according to Pozzebon et al. (25). The Pozzebon et al. model calculates time-dependent film thickness for a grease lubricated rolling bearing where surface roughness effects have also been considered. The Van Zoelen et al. (24) model for pressure-induced side flow is integrated with the Baart et al. (22) centrifugally driven base oil bleed model, and then is combined with the Liu et al. (34) approach of load sharing and friction calculation for mixed-lubricated contact to determine the level of surface interaction and surface roughness effect on the side flow. The film thickness is then used to calculate load sharing between contact asperity and lubricant and provides lubricated friction and wear coefficients. The surface roughness is treated as being constant. Roughness maybe varied over time, particularly during the running-in phase, which may influence the tribological behavior of the bearing (43). Once the running-in phase is completed and the steady-state phase starts (a phase that is considered the most serviceable period of bearing), wear takes place in the mild form and the surface roughness becomes smoother—that is, the variation of roughness over time during this phase is small (43). As bearings usually run for a long time, most of their working life is in the steady-state phase so the assumption of constant surface roughness is justified. The lubricating grease is considered a porous network, formed by the thickener fibers, that contains the base oil. The thickener microstructure has been assumed to be homogeneous anisotropic (i.e., the fibers all tilt perpendicular to the oil flow and pressure gradient). The Baart et al. (22) model has been used for the calculation of grease bleeding. This is based on Darcy's model of viscous flow through a porous thickener microstructure due to a centrifugal force where the bleeding rate depends on pressure gradient, oil viscosity, and permeability of thickener network. However, no tribochemical effects of

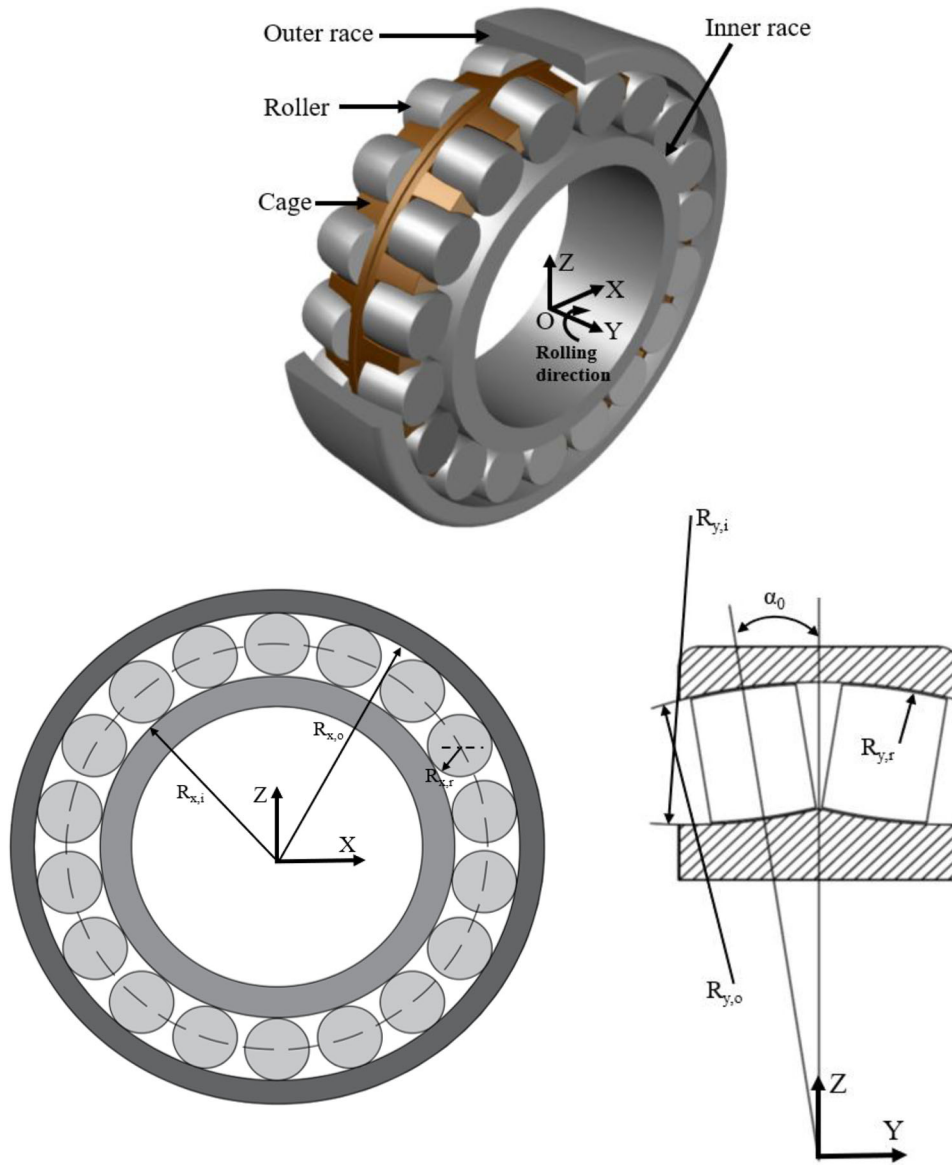


Figure 1. Geometry of a spherical roller bearing.

the grease were addressed in the model. Hence, the grease rheology is the main input for this model.

Figures 2a and 2b show the physical representation and the block diagram of the Pozzebon et al. (25) wear model, respectively. Figure 2a shows the grease reservoir, which is the space available around the cage, between the rollers. However, the grease lubricates both the inner and outer race contact. The Pozzebon et al. model was based on the Baart et al. (22) bleed model, which also predicted base oil supply from the cage reservoirs (laying in the space available around the cage, between the rollers). A similar reservoir concept is considered in the present work. In order to reduce the computational burden, the film thickness is calculated along the central line according to reference (25). The simulation starts with an initial lubricant layer thickness (Step I) that evolves with time and varies only across the contact/race. The model also calculates the roller-raceway contact loads due to an applied radial load and generated centrifugal force (Step II), which is called the internal load distribution. The load experienced by each contact is again

distributed between contact asperity and the lubricant film, and this is called load sharing between asperity and lubricant in this study. The determination of asperity-hydrodynamic load sharing is vital for the friction and wear prediction of mixed lubricated contact, as the asperity component of the contact load contributes to wear only (Step III). Moreover, the load sharing is influenced by the lubricant film/layer thickness, which depends on the lubricant available on the track. The net lubricant available for the next roller contact is from the balance of pressure-induced side-flow loss and supply from the grease reservoir by the centrifugal force (Steps IV to VI). The layer thickness decay rate ($\frac{\partial \tilde{h}_{\infty}(y,t)}{\partial t}$) is described by the summation of rates from the loss ($\frac{\partial \tilde{h}_{L,\infty}(y,t)}{\partial t}$) and supply ($\frac{\partial \tilde{h}_{R,\infty}(y,t)}{\partial t}$) mechanisms as (25)

$$\frac{\partial \tilde{h}_{\infty}(y,t)}{\partial t} = \frac{\partial \tilde{h}_{L,\infty}(y,t)}{\partial t} + \frac{\partial \tilde{h}_{R,\infty}(y,t)}{\partial t} \quad [1]$$

The expression for the central film thickness in the starved rough contact is

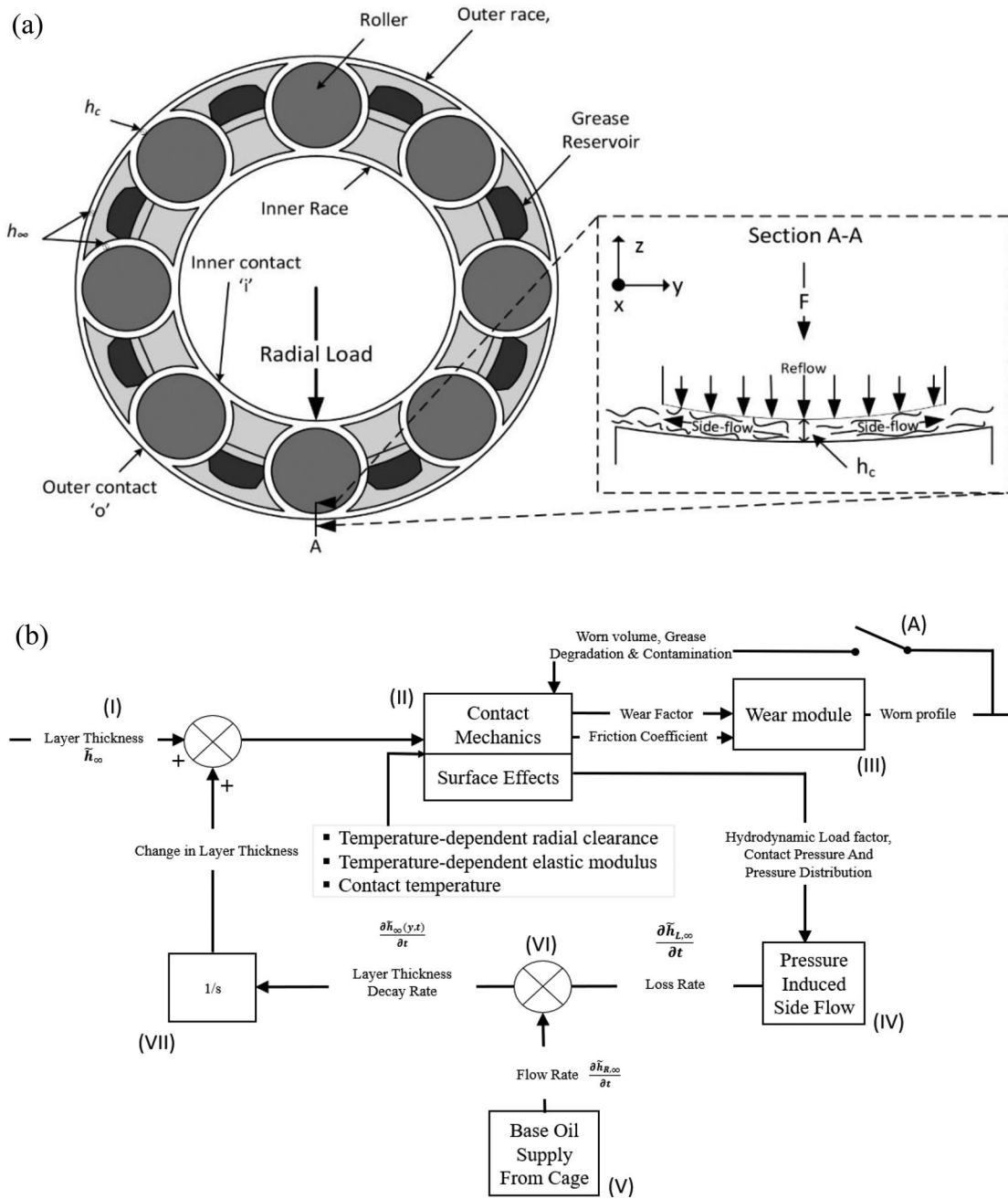


Figure 2. (a) Physical representation and (b) feedback loop of the Pozzebon et al. model [25] for lubricated wear process.

$$h_c = \frac{2\tilde{h}_{oo}}{\bar{p} \left(\frac{F_T(\psi)}{\gamma_1^{1/3} A_{nom}} \right)} \quad [2]$$

where γ_1 is the hydrodynamic load factor and A_{nom} is the nominal contact area.

Finally, the rate of layer thickness is accumulated (Step VII) and added to the layer thickness in Step I. The bleeding rate reduces over time as the oil percentage drops if grease bleeds. In order to reduce the computational time, Pozzebon et al. [25] calculated the bleed rate priorly for the usage period of the model. In this study, the simulation for the total usage period is performed for each temperature, with the grease bleeding rate also varying with temperature. Furthermore, the calculation of internal load distribution, elastic modulus, and contact surface temperature for each bulk temperature is

performed in Step II. The model formulation for temperature-dependent internal load distribution due to applied radial load is presented in the next subsection. The contact patch parameters are also calculated using the elastic modulus under the operating temperature. The temperature-dependent elastic modulus model is presented in the second subsection. The model formulation for contact temperature is presented in the third subsection. The model formulation for friction and wear calculation is presented in the fourth subsection.

Temperature-dependent radial clearance and internal load distribution

In this study, the bearing's working radial clearance is calculated at the operating temperature. If the bearing bulk

temperature rises above ambient temperature, thermal expansion of the bearing components occurs. The radial thermal expansion of the roller ($\varepsilon_{T,r}$) and that of the inner race ($\varepsilon_{T,i}$) are assumed to be linear and calculated as (13, 44)

$$\varepsilon_{T,r} = \alpha_r R_{x,r} T_b, \quad [3]$$

$$\varepsilon_{T,i} = \alpha_i R_{x,i} T_b, \quad [4]$$

where α_r and α_i are the expansion coefficients of the roller and inner race, respectively. $R_{x,r}$ and $R_{x,i}$ are the radius of the roller and the inner race in the rolling direction and are shown in Fig. 1. T_b is the bearing bulk temperature.

The radial expansion of outer-race ($\varepsilon_{T,o}$) is calculated using (13, 44)

$$\varepsilon_{T,o} = \frac{\alpha_o (1 + \nu) R_{x,o}}{3 R_{x,o} + R_h} [T_b (2R_{x,o} + R_h) + T_h (2R_h + R_{x,o})], \quad [5]$$

where α_o are the expansion coefficients of outer race, $R_{x,o}$ is the radius of outer race in the rolling direction as shown in Fig. 1, and R_h is the radius of the housing.

Hence, the total change of bearing radial clearance (ε_T) is

$$\varepsilon_T = \varepsilon_{T,r} + \frac{(\varepsilon_{T,i} - \varepsilon_{T,o})}{2}. \quad [6]$$

By considering the thermal effect, the working radial clearance ($u_{r,w}$) can be calculated as

$$u_{r,w} = u_{r,0} + \varepsilon_T, \quad [7]$$

where $u_{r,0}$ is the initial radial clearance. In this study, the internal load distribution of the radially loaded spherical roller bearing is calculated using the working radial clearance $u_{r,w}$. Further details of the load distribution calculation for the above bearing can be found in Milne and Meehan (45), in which Wan and Zhaoying's (46) iterative method is employed. Here, two important expressions are provided as

$$Q_\psi = Q_0 \left[1 - \frac{1}{2\varepsilon} (1 - \cos \psi) \right]^{1/t}, \quad [8]$$

$$\varepsilon = \frac{1}{2} \left(1 - \frac{u_{r,w}}{2\delta_{\max} + u_{r,w}} \right), \quad [9]$$

where Q_ψ is the roller load when it is positioned at an angle ψ , Q_0 is the roller load when it is positioned at $\psi = 0^\circ$, t is a constant (the value of t is provided in nomenclature), ε is the load distribution factor, and $\delta_{\max}$ is the maximum deformation.

A schematic of the bearing internal load distribution is shown in Fig. 3. The centre position of the load zone ($\psi = 0^\circ$) represents the maximally loaded roller. In the radially loaded bearing, there is a nonloaded zone ($-\psi_e > \psi > \psi_e$) over which the rollers do not share the applied load.

Temperature-dependent elastic modulus and contact patch parameters

The elastic modulus, an important parameter for contact analysis, varies with temperature. The present model considers the change in elastic modulus with temperature in the contact

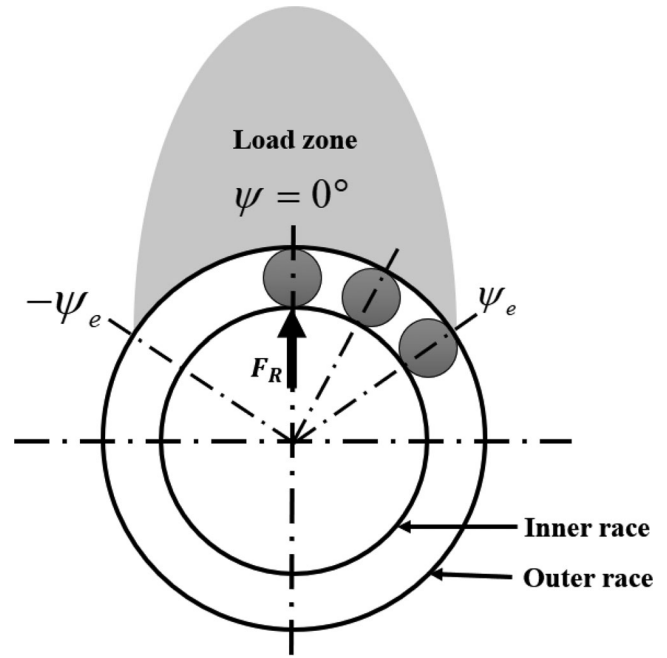


Figure 3. Schematic diagram showing bearing load zone with an upward applied load.

analysis. The elastic modulus of steel is modeled according to Ancaş and Gorbănescu (16). The elastic modulus–temperature relationship used in the present article is for high-carbon steels as bearing steels. Ancaş and Gorbănescu's (16) expressions of temperature-dependent elastic modulus are given as

$$E(T_b) = E(20) \left(1.0 + \frac{T_b}{2000 \ln(T_b/1100)} \right) \quad 20^\circ\text{C} < T_b < 600^\circ\text{C},$$

$$E(T_b) = E(20) \left(\frac{690 - 0.69T_b}{T_b - 53.5} \right) \quad 600^\circ\text{C} < T_b < 1000^\circ\text{C}, \quad [11]$$

where $E(20)$ and $E(T_b)$ are the elastic modulus at 20°C and T_b , respectively. The value of $E(20)$ can be found in the nomenclature.

The Hertzian contact dimensions a (half contact length in the rolling direction) and b (half contact length in the transverse direction), and the maximum and mean contact pressure p_h and p_{mean} of each roller–raceway contact depend on the corresponding roller load that also varies with the roller azimuth position ψ in the rolling bearing and the elastic modulus. The expression of these parameters for the bearing contact can be given as (24)

$$a(\psi, T_b) = \left(\frac{6R'\zeta\chi}{E'(T_b)\pi} F_T(\psi) \right)^{1/3}, \quad [12]$$

$$b(\psi, T_b) = \frac{a(\psi, T_b)}{\zeta}, \quad [13]$$

$$p_h(\psi, T_b) = \frac{3F_T(\psi)}{2\pi a(\psi, T_b)b(\psi, T_b)}, \quad \text{and} \quad [14]$$

$$p_{\text{mean}}(\psi, T_b) = \frac{F_T(\psi)}{\pi a(\psi, T_b)b(\psi, T_b)}, \quad [15]$$

where χ and ζ are the elliptic integral of a first and second kind respectively; F_T is the total contact load; E' is equivalent elastic modulus; and R' is the reduced radius of the contact.

Contact surface temperature

The contact temperature (T_c) of a rolling-sliding contact is the sum of the bulk temperature (T_b) and the temperature rise (ΔT),

$$T_c = T_b + \Delta T. \quad [16]$$

Masjedi and Khonsari (32) used Tian and Kennedy's analytical expression (47) to estimate the temperature rise in the oil lubricated mixed-EHL contact where both fluid shear heating and asperity friction are considered as the heat source. This expression has been considered in the present study of grease-lubricated contact as there is no published expression specifically for grease so far. This study also considers both lubricant shear heating and asperity friction as the total heat source to treat the roller-race contact, assuming the contact is being lubricated in the mixed-EHL regime. The expression of flash temperature rise (ΔT) according to Tian and Kennedy (47) is

$$\Delta T = \frac{2aq}{\sqrt{\pi}(k_1\sqrt{1+Pe_1} + k_2\sqrt{1+Pe_2})}, \quad [17]$$

where q is the average heat flux, and $k_{1,2}$ and $Pe_{1,2}$ are the thermal conductivities and the Peclet numbers of contacting surfaces, respectively. The expression of Peclet number Pe is $Pe = u_s a \rho C_p / (2k)$. In this study, the contact surface temperature calculated is the average one, which is for the entire range of Peclet number, according to Tian and Kennedy (47). Hence, the Peclet number is calculated using the average sliding speed, $u_s = \omega_r R_{r,x} \xi_{avg}$ (24). Here, ξ_{avg} is the average of the absolute creep ratio across the contact patch. The creep ratio is the normalized relative velocity, which varies across the contact patch due to the variation of the local rolling radius of the roller. Hence, an average is used to calculate Peclet number. More information about the creep ratio of bearing contact can be found in references (45) and (48). In this study, AISI 52100 bearing steel was considered and its corresponding specific heat, C_p , and thermal conductivity, k , are used (49), as provided in the nomenclature as well. In this work, for the mixed lubricated condition, the heat flux, q , is assumed as the sum of frictional heat q_A (generated by the asperity interaction) and q_H (generated by the lubricant shear):

$$q = q_A + q_H \quad [18]$$

The asperity component of heat flux, q_A , can be approximated as

$$q_A = \mu_A u_s p_A, \quad [19]$$

where μ_A is the asperity component of friction coefficient and p_A is the asperity part of the contact pressure.

The hydrodynamic component of the heat flux q_H can be approximated using the lubricant's mean shear stress ($\tau_{H,mean}$) and sliding speed (u_s) (32, 50),

$$q_H = \tau_{H,mean} u_s. \quad [20]$$

In this study, the mean hydrodynamic shear stress ($\tau_{H,mean}$) is calculated from the Eyring shear stress model by assuming that the contact is operated at a mean high constant pressure and viscosity is varied with pressure according to the Barus equation as obtained in Spikes and Jie (50). The Eyring shear stress equation is

$$\tau_H = \tau_0 \arcsin h \left(\frac{\eta \dot{\gamma}}{\tau_0} \right), \quad [21]$$

where τ_H is the lubricant shear stress, τ_0 is the Eyring shear stress, $\dot{\gamma} = u_s/h_c$ is the shear strain rate, and η is the base oil viscosity. Here, h_c is the central film thickness. The Eyring shear stress equation was developed mainly for oil lubricated contact, and its use for grease lubricated contact in this study may provide an approximate result. The mean lubricant shear stress ($\tau_{H,mean}$) is calculated using the mean hydrodynamic pressure $p_{H,mean}$ (50, 51) as

$$\tau_{H,mean} = \tau_0 \left(\log_e(\dot{\gamma}) + \log_e \left(\frac{2\eta_0}{\tau_0} \right) + \alpha p_{H,mean} \right), \quad [22]$$

where η_0 and α are viscosity at atmospheric pressure and pressure-viscosity coefficient, respectively. In the heat flux calculation, the rolling resistance loss is not considered, as the rolling friction coefficient is usually very small (<0.02 for steel-steel contact (52, 53)) when compared to the sliding friction coefficient.

In mixed EHL contact, the total contact load is shared by contact asperities and lubricant. Johnson et al. (54) first proposed an approach to determine the load distribution between contact asperities and lubricant. The total contact load (F_T) can be written as the sum of the load carried by contact asperities and lubricants (34, 54, 55),

$$F_T = F_A + F_H \quad [23]$$

where F_H and F_A are the hydrodynamic and asperity portion of the contact load. Similarly, the total contact pressure (p_T) is also composed of two components, asperity pressure (p_A) and hydrodynamic pressure (p_H), and can be written as

$$p_T = p_A + p_H. \quad [24]$$

The load sharing is usually defined by two load factors (34), γ_H (hydrodynamic) and γ_A (asperity):

$$\gamma_H = \frac{F_T}{F_H} = \frac{p_T}{p_H}; \quad \gamma_A = \frac{F_T}{F_A} = \frac{p_T}{p_A}; \quad [25]$$

The load sharing holds the condition

$$\frac{1}{\gamma_A} + \frac{1}{\gamma_H} = 1. \quad [26]$$

In this study, the load sharing is determined according to the Liu et. al. (34) approach. Note that the preceding approach of the load-sharing calculation is widely used for oil-lubricated rough (mixed-lubricated) contact. The grease-lubricated contact is mostly run under mixed-lubrication conditions in which both the lubricant and asperity interactions take place (31). The approach should be useful for grease-lubricated rough contact as well, because it is an iterative method, and in

grease-lubricated rough contact, both grease films and surface asperities share the total contact load as occurs with oil-lubricated contact (31). However, further verification for grease lubricants is required.

Friction and wear model

The friction coefficient (μ) of a mixed lubricated contact is composed of two components: the asperity component (μ_A) and the hydrodynamic component (μ_H). In this study, the friction coefficient is calculated according to Liu et al. (34) as

$$\mu(\psi, \tilde{h}_\infty) = \underbrace{\frac{\mu_b F_A(\psi, \tilde{h}_\infty)}{F_T(\psi)}}_{\mu_A} + \underbrace{\frac{\tau_H(\psi, \tilde{h}_\infty) A_H(\psi, \tilde{h}_\infty)}{F_T(\psi)}}_{\mu_H} \quad [27]$$

where τ_H is the hydrodynamic shear stress and A_H is the hydrodynamic part of the contact area. A_H is calculated using the hydrodynamic load factor γ_H and the nominal contact area A_{nom} as $A_H = (1/\gamma_H)^{2/3} A_{nom}$ (34). τ_H is calculated using the Eyring shear stress equation as shown in Eq. [21], in which the temperature-dependent lubricant's viscosity is calculated using Roeland's equation (34,56) as

$$\eta(p_H, T_c) = \exp \left[\ln(\eta_0) + 9.67 \left(1 + \frac{p_H}{1.962 \times 10^8} \right)^{\alpha^*} \times \left(\frac{T_a + 135}{T_c + 135} \right)^{s_0} - 9.67 \right] \quad [28]$$

where $\alpha^* = \alpha \frac{1.962 \times 10^8}{\ln(\eta_0) + 9.67}$, and s_0 is the viscosity-temperature coefficient. The values of α and s_0 are provided in the nomenclature.

As already stated, in a mixed-EHL contact, the total contact load is shared by both lubricants and contact asperities. The portion of contact load carried by lubricant does not contribute to the wear process, whereas wear is mainly contributed by the portion of contact load that is carried by contact asperities (37, 55). Therefore, in the calculation of wear, only the asperity portion (F_A) of the contact load (F_T) is an input into the wear equation. The wear is calculated using the frictional power wear equation (25,45) as

$$\Delta w_{i/o}(y, \psi, \tilde{h}_\infty, T) = \frac{K}{\rho_s} \left| f_x(y, \psi, \mu_A(\psi, \tilde{h}_\infty, T)) \right| \xi(y, \psi, \mu(\psi, \tilde{h}_\infty, T)), \quad [29]$$

where K is the boundary frictional power wear coefficient, ρ_s the density of bearing material, f_x the traction in the rolling direction, and ξ is the creep ratio.

The model's wear prediction at some bearing location (y, ψ) for one revolution is given by $w_{i/o}(y, \psi)$, where the subscript i stands for inner race contact and o stands for outer race contact, so the mean local wear profile change for one roller's revolution about the shaft can be described as

$$\Delta W_{R,A}(y) = \text{mean}_{i/o}(w_{i/o}(y, \psi)), \quad -\pi \leq \psi \leq \pi. \quad [30]$$

If there are Z_r rollers on each side of the bearing and the shaft rotation rate differs from the cage, the mean local wear profile change for a given number of shaft rotations (n) is

$$W_{I,A}(y, n) = n Z_r R_p \Delta W_{R,A}(y), \quad [31]$$

where R_p is the ratio of cage rotation per input shaft rotations, and according to the equivalent planetary gearbox ratio this is

$$R_p = \frac{R_{x,i}}{2(R_{x,i} + R_{x,r})}. \quad [32]$$

Finally, the volume material loss, V , of the spherical roller bearing for a given number of shaft rotations (n) is (25)

$$V(n) = \pi \int_{\text{RollerWidth}} \left(\left(W_{I,A}(y, n) - R_{x,i} \right)^2 - R_{x,i}^2 + \left(W_{I,A}(y, n) - R_{x,o} \right)^2 - R_{x,o}^2 \right) dy. \quad [33]$$

Numerical solution

The model is implemented in MATLAB and the process flow of the model is from Pozzebon et al. (25), as shown in Fig. 4. The simulation starts by precalculating the bleed rate for the usage period of the model. This is done by taking the base oil supply model with no feedback mechanisms. This section illustrates the integration of each component of the present model with the process flow of Pozzebon's model. The change of radial clearance and elastic modulus with temperature were two major additions in the present work. These two components have been precalculated (Eqs. [7] and [10]) and used as input for the rest of the program. Another addition was the calculation of contact surface temperature, which is the sum of the bulk temperature and average flash temperature rise in the contact surface (Eq. [16]). To determine the flash temperature rise, it is necessary to calculate the heat flux of the contact, which is composed of asperity and hydrodynamic components. Thus, the load sharing between the contact asperity and lubricant film needs to be calculated first, and this is performed at the bulk temperature. Then the flash temperature rise is calculated (Eq. [17]) and added to the bulk temperature to obtain the contact surface temperature (Eq. [16]). In this study, the simulation is performed in 20 steps using a bearing load of 20 kN and speed of 480 rpm. The roller-race contact is a curved contact. For the robustness of the model, the film thickness is calculated along $y=0$ (25). This is justified by the results presented by Van Zoelen et al. (24), who showed that the film thickness in starved lubricated cases becomes increasingly constant across the contact, and by Huang et al. (57), who showed this experimentally for point contact cases with minimal reflow. This made it possible to avoid the calculation of load sharing for more than one film thickness in any given contact (25). The ordinary differential equation of central film thickness was solved numerically using Euler's method. For the wear volume, Eq. [34] is solved using a trapezoidal integration rule. Using a desktop

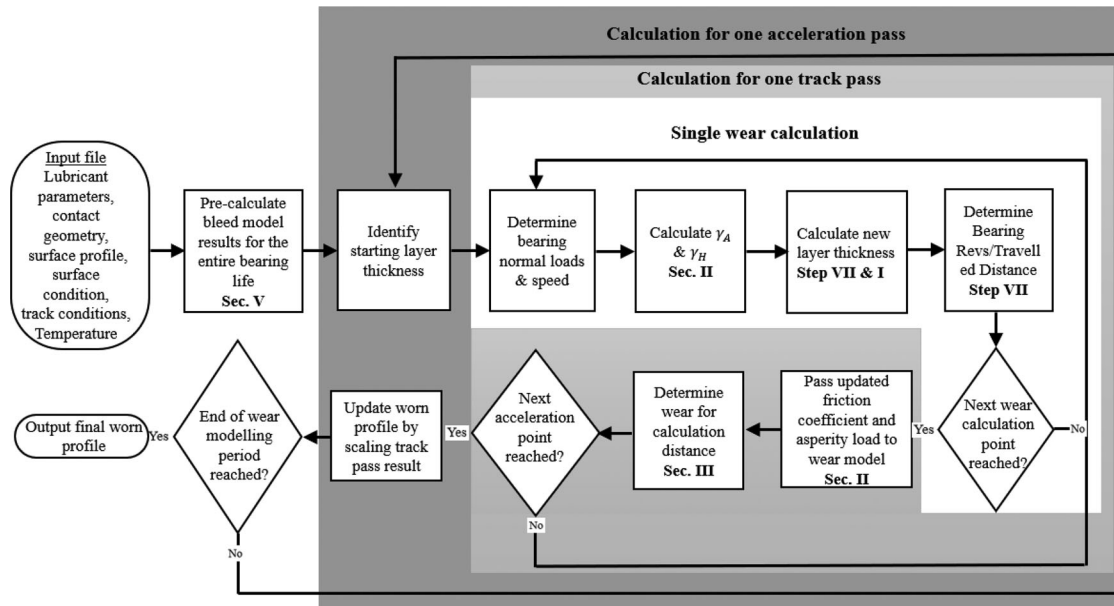


Figure 4. Schematic diagram of the Pozzebon et al. (25) model.

computer of I7-8700 CPU and 16 GB of ram, each run takes approximately 12–16 minutes. The input parameters and the corresponding values are provided in the nomenclature.

Results and discussion

The model is used to predict the wear of a grease-lubricated spherical roller bearing by considering the temperature effect. The simulation is performed using a double-row spherical roller bearing with 22 rollers on each side. The main parameters of the bearing are provided in Table 1. The wear results are produced for 200 million bearing revolutions. The boundary friction and wear coefficients were obtained experimentally by Pozzebon et al. (25) and are provided in Table 2. To account for the effect of wear on these coefficients, they are interpolated for each wear step based on the wear volume as per Pozzebon et al. (25). The important grease properties for the model input are provided in the nomenclature. The results are presented and discussed in the following sections.

The effect of temperature on the elastic modulus and contact parameters

The elastic modulus of steel is known to reduce with the increase in temperature. A graph showing the elastic modulus change with temperature for steel is reproduced from reference (58), as shown in Fig. 5a. A comparison of the change of elastic modulus of different steels with temperature can be found in reference (58). When the temperature was less than 200 °C, the difference was small (approximately 2–3%). However, if the temperature is much higher (higher than 400 °C), the elastic modulus variation between different steels increases to approximately 10%. Thus, the equations used in this study can be applied to a wide variety of steels if the temperature is not raised to an extremely high value. Furthermore, the contact patch dimensions (the contact patch dimensions (a and b) and mean contact pressure (P_{mean})) of both the inner and outer

Table 1. The basic bearing parameters.

Parameters	Value
Radius of roller in the rolling direction, $R_{x,r}$ (m)	9.9×10^{-3}
Inner race radius in the rolling direction, $R_{x,i}$ (m)	82.7×10^{-3}
Outer race radius in the rolling direction, $R_{x,o}$ (m)	-102.4×10^{-3}
Radius of roller in the transverse direction, $R_{y,r}$ (m)	100.3×10^{-3}
Inner race radius in the transverse direction, $R_{y,i}$ (m)	-103×10^{-3}
Outer race radius in the transverse direction, $R_{y,o}$ (m)	-102.4×10^{-3}

Table 2. Boundary wear and friction coefficients (25).

Reference usage distance (km)	0	5000,000	1,800,000
Wear volume (mm^3)	0	1.920	11.595
Friction coefficient	0.151	0.147	0.212
Wear coefficient (kg/N.mm)	7.8×10^{-17}	1.189×10^{-16}	1.262×10^{-16}

contact of the bearing are calculated by considering the temperature effect on the elastic modulus. The results are plotted in Fig. 5b. Here, a is the maximum half-contact length in the rolling direction, and b is the maximum half-contact length in the transverse direction. The calculations are performed by varying the temperature only, while the rest of the other parameters are constant. For the temperature range considered, the decrease of the elastic modulus is not as drastic and is only approximately 3%. As the contact patch dimensions (a and b) are inversely related to the elastic modulus, the reduction of the elastic modulus increases both a and b . For a temperature rise from ambient (20 °C) to 110 °C, the increase of a and b was no more than 1% in both the inner and outer contact. Moreover, this small increase in contact patch dimensions causes a slight decrease in the mean contact pressure (less than 2%). Therefore, this range of temperature appears to have only a minor effect on contact patch dimensions and pressures.

The effect of temperature on the internal load distribution

The radial clearance of the bearing is usually changed during operation due to the thermal expansion of the bearing

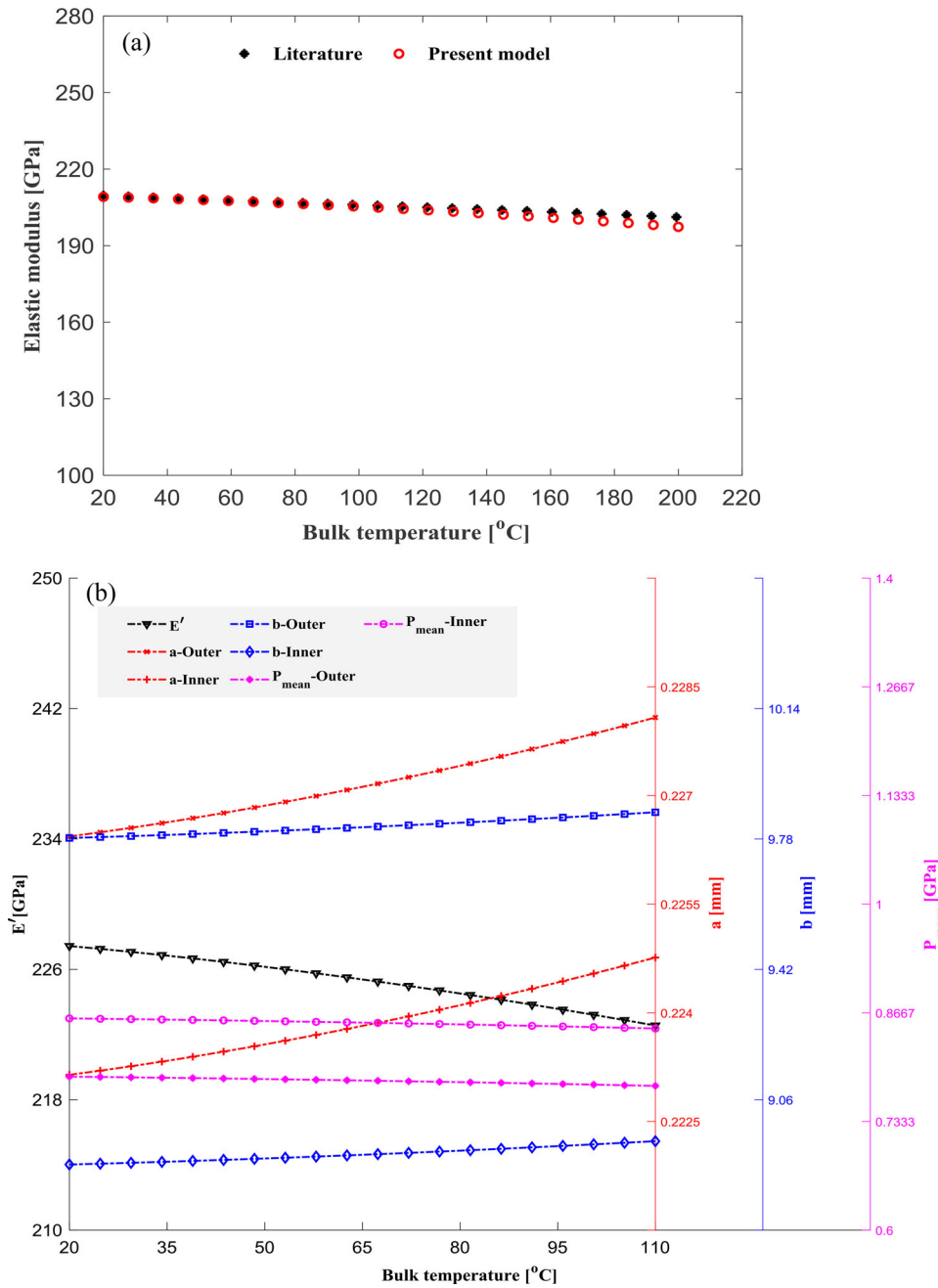


Figure 5. (a) Comparison of elastic modulus against temperature between the present model and experimental result from reference (58); and (b) elastic modulus and corresponding semicontact lengths (a and b) and mean pressure of inner/outer contact of maximally loaded roller ($\psi = 0^\circ$) of the spherical roller bearing.

components. The thermal expansion is calculated using the model as shown in the second section. The housing radius is equivalent to the outer radius of the bearing. Initially, the temperature-dependent radial clearance model is validated by comparing using results predicted by Bian et al. (12) as shown in Fig. 6. Note that Fig. 6 shows only the change in the amount of radial clearance, and the following paragraph presents the radial clearance and the internal load distribution of the double-row spherical roller bearing due to the temperature variation.

The present study investigates the temperature effect on radial clearance by increasing it from ambient (20°C) to 110°C. It is assumed that the temperature of the whole bearing varies from one isothermal state to another isothermal state, so no temperature variation between bearing

components is considered. The working radial clearance is presented in Fig. 7a. Note that the initial radial clearance is 190 μm , which is assumed to be the radial clearance for ambient temperature. For a temperature rise from ambient to 110°C, the radial clearance is reduced by 12%, which further causes a change in the internal load distribution inside the bearing. As stated before, the contact loads distributed inside a bearing due to the applied radial force are a function of radial clearance (9). Figures 7b and 7c show the load distribution of the inner and outer contact, respectively, for different temperatures. The roller position around the races is plotted on the horizontal axis and the roller load (or contact force) on the vertical axis. Here, the intention is to present the load distribution mainly; therefore, only $\psi = 45^\circ$ to each side from roller position $\psi = 0^\circ$ is shown in the figure.

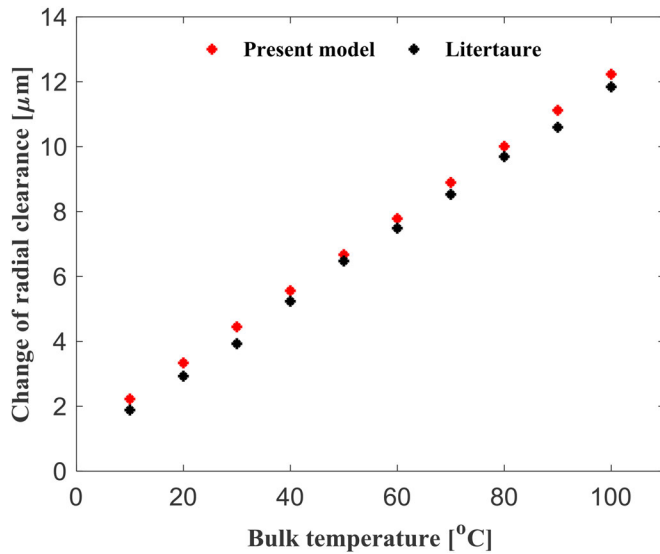


Figure 6. Comparison of change of radial clearance with temperature between this article and data available in reference (12).

The data plotted in each curve represented the roller load at the respective roller positions when one of the rollers is assumed to be positioned at $\psi = 0^\circ$. In all cases, the load distribution resembles a bell curve and is symmetric about the $\psi = 0^\circ$ roller position. Figures 7b and 7c show that increasing temperature for a given bearing radial load causes the roller load at the central zone to decrease and the load zone, and the size of the races that mainly support the load, to increase. For the temperature rise from ambient to 110°C , the load for the $\psi = 0^\circ$ roller position is decreased by approximately 4%, whereas the load zone is increased by about 5%. This indicates that for each revolution of a roller around the races, the roller is in contact with the races for a longer distance. This change of load distribution is due to the decrease of radial clearance with the temperature rise (59), as shown in Fig. 7a. Hence, with the decrease of radial clearance, rollers contact the races for a longer distance at the two ends of the loaded zone, and this causes the loaded zone to increase and the rollers at the central zone to carry less load (9, 60). Therefore, the change of load distribution with temperature is attributed to the change of bearing radial clearance due to the thermal expansion of the bearing.

The film thickness and lubrication regime analysis

The wear of a lubricated rough bearing largely depends on the ability of a lubricant film to separate the rolling elements and the raceways as this film thickness determines the load sharing between contact asperity and lubricant. Figure 8a (left axis) shows the central film thickness for a temperature rise from 50 to 110°C . The film thickness decreases with the temperature rise due to its effect on viscosity. A plot of dynamic viscosity change with temperature is also shown in Fig. 8b. In the present model, the equation of film thickness is formulated for the rolling element bearing geometry by considering the combined effect of pressure-driven loss and centrifugally driven supply, which is much different from the usual consideration of a ball-on-disc single contact

system for the experimental measurement of film thickness. A comparison of the percentage change of central film thickness between the present work and the experimental data available in Gonçalves et al. (61) for a grease-lubricated ball-on-disc contact is shown in Fig. 8c. In the Gonçalves et al. (61) study, the film thickness was measured using a space layer interferometry method. At the ball-on-disc contact was directed high-intensity light, which was divided into two: a part reflected from the underside of the disc to the spectrometer, and another part passed through lubricant film and reflected back from the ball surface, producing interference between two light beams. The interference of two light beams that have traveled different distances produces an interference image from which the film thickness was computed. The literature (61) measured film thickness for three temperatures: 40, 80, and 110°C . The data used in this article are for 0.5% of speed; however, the trends of graphs from 0.1 and 1% were quite similar. Both graphs show a decreasing trend of film thickness with temperature rise. This confirmed the model prediction that increasing the temperature reduces viscosity and therefore the film thickness. The offset found between the two graphs is likely due to the different geometry of the spherical roller bearing compared to the simplified experimental ball-on-disc. In the single contact system, like in the Gonçalves et al. (61) study, the time difference between successive over-rollings in the ball-on-disc configuration is large and simulation of lubricant flow and the feed and loss mechanisms is almost impossible (20). Moreover, the centrifugal forces that act on the grease in rolling-element bearings cannot be simulated in the ball-on-disc devices. Experimental measurement of the film thickness in a spherical roller bearing is challenging at present but could be an area of future research.

The lubrication regime is usually represented with the lambda (λ) ratio, a ratio of lubricant film thickness to composite surface roughness (19, 61),

$$\lambda = \frac{h_c}{\sqrt{\sigma_r^2 + \sigma_{i/o}^2}}, \quad [34]$$

where h_c is the central film thickness and σ_r and $\sigma_{i/o}$ are the mean roughness height of the roller and inner/outer races, respectively. Usually, λ is calculated using the minimum film thickness and root mean square (RMS) roughness (62); however, this study calculated it using the central film thickness and mean roughness height of the contacting surfaces. Generally, $\lambda > 3$ refers to surfaces that are completely separated by the lubricant film, $3 \geq \lambda \geq 1$ indicates the contact is under mixed lubricated conditions, which means that lubricant film separates the contact surfaces but a frequent metal-to-metal contact occurs, and $\lambda < 1$ means the load is mainly supported by the contact asperities. A transition between lubrication regimes can be obtained by changing either the roughness or the central film thickness. Surface roughness always plays a significant role in the frictional and wear behavior of a lubricated contact. In this work, the surface roughness is assumed to be constant as mentioned earlier and the λ value mainly changes with the central film thickness. The value of λ is plotted in Fig. 8a

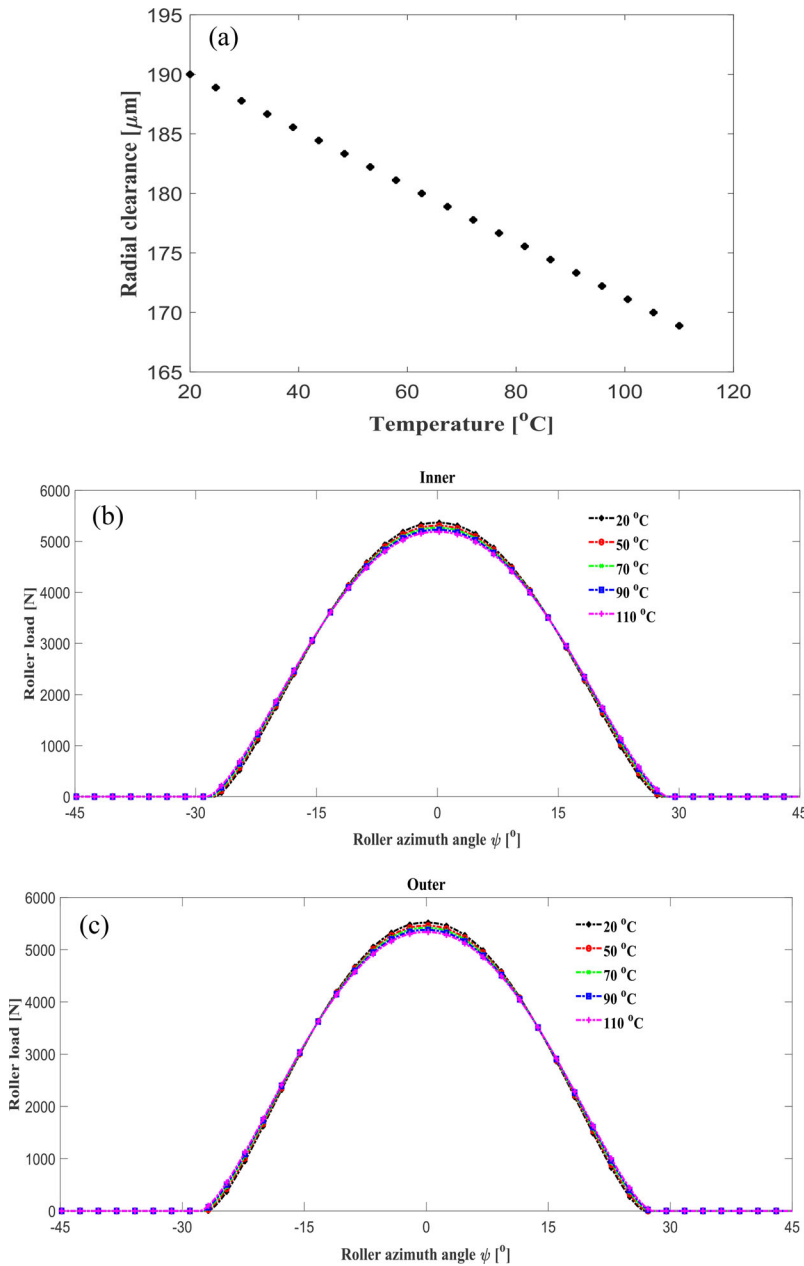


Figure 7. (a) Working radial clearance; (b) the internal load distribution of the inner and (c) of the outer contact of a double row spherical roller bearing at temperatures 20, 50, 70, 90, and 110 °C.

(right axis) as a function of temperature. The λ ratio decreases with temperature rise due to the decrease in film thickness. When the temperature reaches 80 °C, λ becomes smaller than 1 (Fig. 8a), indicating that most of the contact loads are carried by contact asperities. However, the literature (62) also shows that the full film and mixed lubrication are possible even at a smaller value of lambda ($\lambda < 1$), meaning that the contact loads still can be carried by the lubricant film, as shown later.

The load sharing between contact asperity and lubricant

In the present study, the internal load distribution described earlier is defined as the load supported by the roller-inner/

outer race contacts due to the radially applied load. On the other hand, load sharing is defined as the portion of load shared between the contact asperities and lubricants of a roller-race contact. Load sharing is also called load distribution between the lubricant and contact asperities in the literature, but in this study, asperity- and lubricant-carried loads are always called load sharing to remove the confusion for readers. The load-sharing calculation is important to predict the wear of a lubricated rough bearing, as the asperity component of the contact load mainly contributes to wear. Figure 9 shows the plot of asperity-carried (F_A) and lubricant-carried (F_H) load of both inner and outer contact for $\psi = 0^\circ$ against temperature. These results are produced without considering the change in internal load distribution due to temperature to see the sole effect of temperature on the load sharing. Moreover, the figure also shows a total

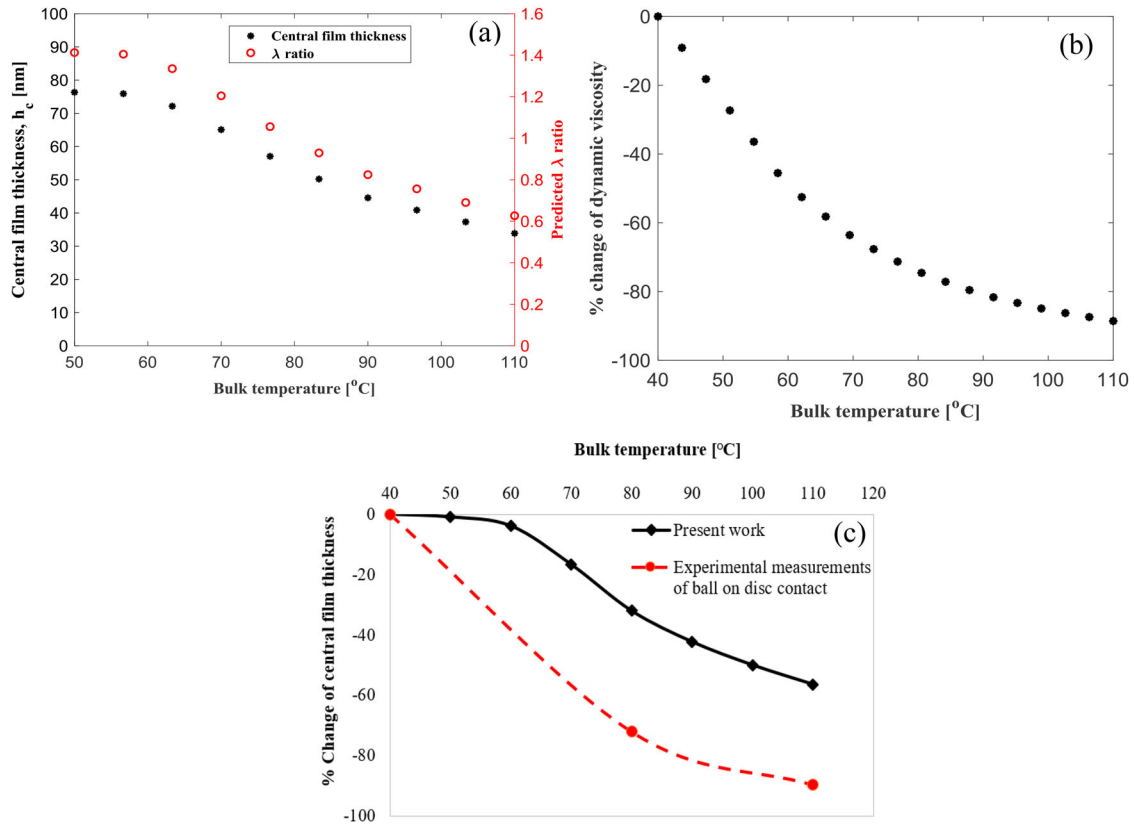


Figure 8. The change of (a) central film thickness and λ ratio for maximally loaded position, (b) percent change of dynamic viscosity with temperature, and (c) a comparison of percent change of central film thickness with data extracted from Gonçalves et al. (61).

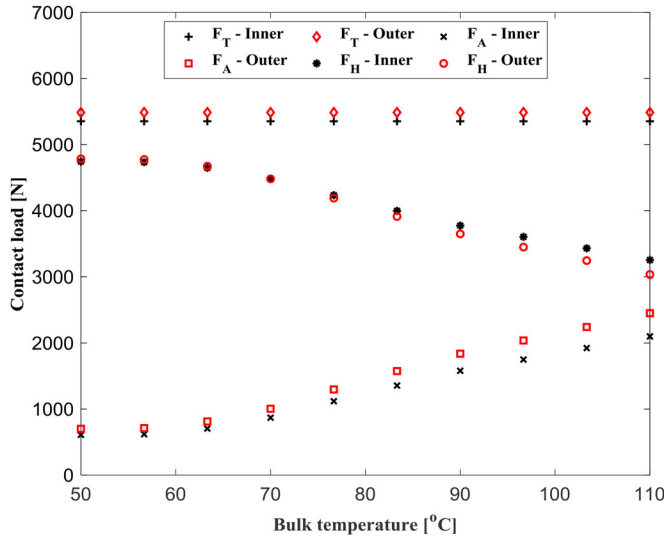


Figure 9. The asperity- and lubricant-carried load of both the inner and outer contact for the maximally loaded contact position ($\psi = 0^{\circ}$). F_T is total contact load, F_A is asperity portion of the contact load, and F_H is hydrodynamic portion of the contact load.

load (F_T) of both inner and outer contact for the roller $\psi = 0^{\circ}$. As can be seen in the figure, at low temperature, most of the contact load is carried by the lubricant, which decreases significantly when the temperature is increased. Similar behavior was also observed by Zhao et al. (40) for their Cu-based friction pair. In the present study, as the temperature is increased from 50 to 110 $^{\circ}\text{C}$, F_A increases by approximately 350%. The increase of F_A with temperature is

attributed to the reduction of lubricant film thickness as discussed in the previous section, which causes an increase in the metal-to-metal contact. As can be seen in the graph, at higher temperatures (above approximately 80 $^{\circ}\text{C}$), F_A is increased at a slightly higher rate. However, the trend of the change of F_A is similar to that of the film thickness. In the previous section, it is shown that the λ value, an indicator of lubrication regime, is less than 1 if the temperature reaches 80 $^{\circ}\text{C}$, at which almost all the contact load is carried by the contact asperity. However, according to Hansen et al. (62), mixed lubrication, a lubrication condition at which both the lubricant and asperity share the contact load, is also possible at a lambda value less than 1. Figure 9 also shows that the contact load carried by lubricant is considerable at a temperature greater than 80 $^{\circ}\text{C}$.

The friction and wear of the bearing

This study predicts the wear of a spherical roller bearing due to the sliding present in roller-raceway contacts. In a spherical roller bearing, almost all the roller-raceway contact is in sliding motion, and that causes a mild form of wear if metal-to-metal contacts cannot be avoided except for the two pure rolling lines (48). The present study focuses on the prediction of mild wear, assuming mixed-EHL contact in which lubricant and contact asperities coexist and share the contact load. The wear is contributed by the asperity portion of the contact load only (i.e., metal-to-metal contact portion), which also depends on the lubricant performance

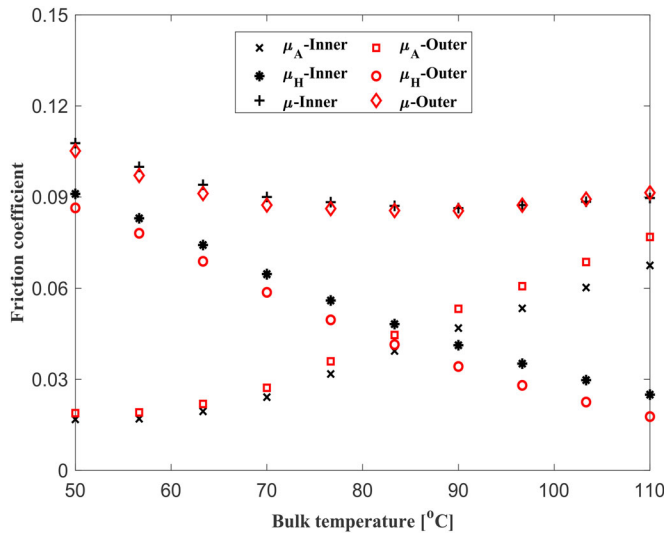


Figure 10. Friction coefficient of both inner/outer contacts against temperature when $\psi = 0^\circ$. [μ , lubricated friction coefficient, μ_A , asperity component of μ , and μ_H , hydrodynamic component of μ].

(63). In this study, the metal-to-metal contact load is predicted first for various temperatures and then used to predict wear. This section presents and discusses the friction and wear results of the grease-lubricated spherical roller bearing under different temperatures. The temperature can influence both rheological properties and the tribochemical behavior of grease, which can further affect the frictional and wear behavior of the contact. In this work, the friction and wear are investigated by considering the temperature effect on the rheological properties. Thus, the tribochemical effect of grease on friction and wear will not be discussed here. Some of the tribochemical phenomena of the grease-lubricated spherical roller bearings on a rail axle can be found in Lin and Meehan's paper (64, 65) in which they performed the elemental analysis of wear particles and microstructure characterization of the degraded grease.

The friction coefficient is an indicator of energy loss, and that is an indicator for wear as well. In a tribopair, a higher friction causes more material loss (31). Figure 10 shows a plot of the friction coefficients (μ : mixed-lubricated friction coefficient, μ_A : asperity component of the friction coefficient, and μ_H : hydrodynamic component of the friction coefficient) of both inner and outer contact for the maximally loaded contact position ($\psi = 0^\circ$) against temperature variation. The trend of the graphs shown in Fig. 10 indicates that at a lower temperature the contact is dominated by the hydrodynamic component of the friction coefficient, μ_H , and at high temperatures, the asperity component of the friction coefficient, μ_A , dominates the contact over the hydrodynamic component. When the temperature is increased from 50 to 110°C, μ_A becomes more than three times higher; however, a corresponding drop of μ_H is also found. The increase of μ_A is attributed to the increase of F_A with temperature, as shown in Fig. 9, as μ_A is scaled using F_A (Eq. [27]). The friction coefficient μ , as shown in Fig. 10, is composed of the asperity (μ_A) and hydrodynamic (μ_H) components; μ is decreased initially until the temperature is approximately 80°C and then starts to increase again. The

graph of μ_A shows that the increasing rate is higher at higher temperatures, which is the reason why μ increases at high temperature. The boundary friction coefficient μ_b is used to scale μ_A . If there is no hydrodynamic component, μ is equivalent to μ_b . As discussed before, the λ value is less than 1 at temperatures greater than 80°C, and the lubricant still carries a portion of the contact load, as shown in Fig. 9. As a result, the hydrodynamic component still contributes to the friction coefficient and μ is much less than the boundary friction coefficient (approximately 0.09). However, μ would approach the boundary value if the portion of load that is carried by the lubricant were zero.

The creep and traction are two major inputs in the wear calculation which are presented for various temperatures. The mean creep ratio (ξ_x) of both inner/outer contact for the maximally loaded roller position ($\psi = 0^\circ$) against temperature is plotted in Fig. 11a. The term ξ_x is defined as the average absolute creep ratio across the contact patch and the result shows no change in it with temperature. The asperity component of the traction f_x of both inner and outer contact for the maximally loaded roller position is also plotted against temperature in Fig. 11b and as a percentage of the total traction of the contact in Fig. 11c. The asperity portion of traction increases with temperature. If the temperature is 50°C, only approximately 20% of the total traction is contributed by the contact asperity. This is almost 80% when the temperature is increased to 110°C. For the maximally loaded roller position, if the temperature is increased from 50 to 110°C, the asperity component of traction $f_{x,A}$ increases from 86.3 to 344.4 N in the inner and from 100.4 to 405.5 N in the outer contact; see Fig. 11c. The increase of traction with temperature is also associated with the increase of the asperity component of the contact force.

Figure 12 shows the wear volume loss (mm^3) of a grease-lubricated spherical roller bearing for the temperature range of 50–110°C. The figure also shows the wear volume loss when the change of internal load distribution and elastic modulus with temperature are considered. As can be seen in Fig. 12, the wear loss increases nonlinearly with temperature. This is primarily due to the large reduction of lubricant film thickness due to temperature rise, which is associated with the rheological properties changing. Specifically, the reduction of viscosity with temperature rise weakens the hydrodynamic film's action, increasing the asperity component of the contact load. In lubricated rough contact, the wear is mainly contributed by the asperity component of the contact force, and it is increased with temperature, as shown in Fig. 9. For the first case when no change of the internal load distribution and elastic modulus with temperature is considered, the wear volume loss increases by 238% for the bulk temperature rise from 50 to 110°C. If the internal load distribution is considered, the predicted wear volume is only 4% less (234%). It was observed that the reduction in elastic modulus and the change of internal load distribution with the temperature rise do not influence the wear prediction significantly. The decrease in the elastic modulus caused an increase in the contact area and a corresponding drop of mean contact pressure, as shown in Fig. 5. The inverse relation between contact pressure and

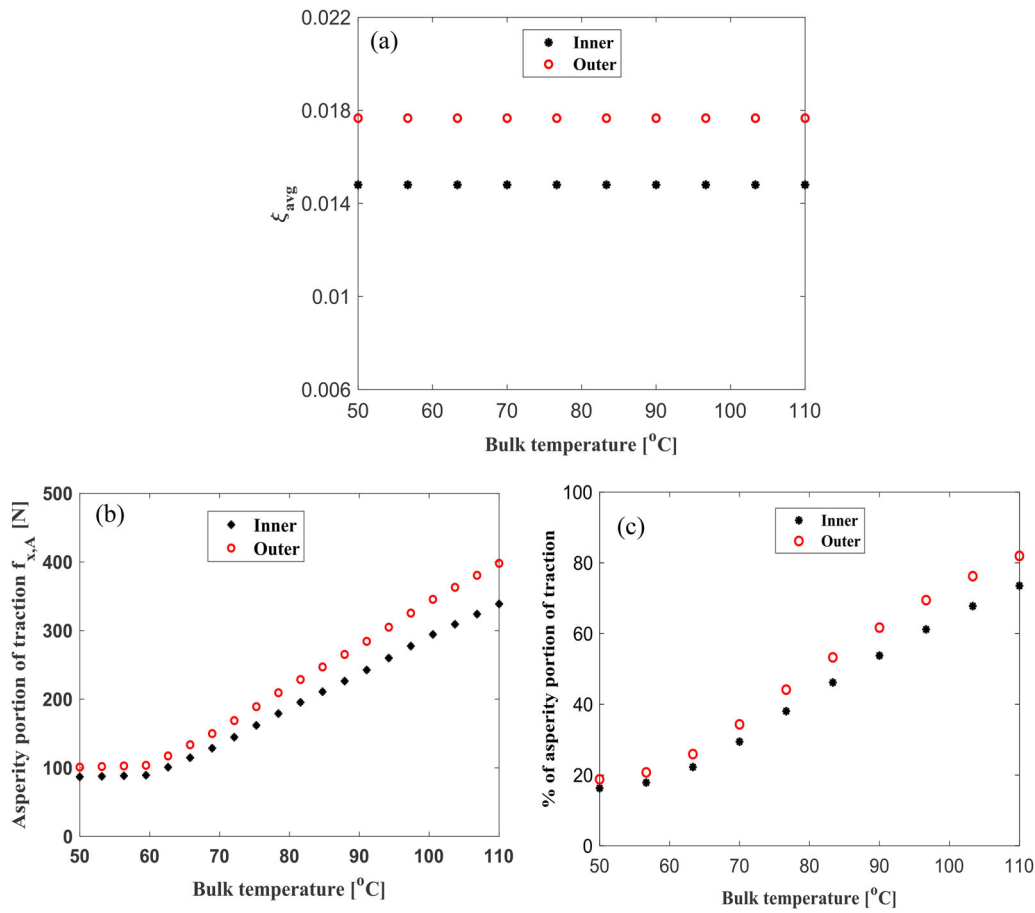


Figure 11. Effect of temperature on (a) mean creep ratio, (b) asperity component of traction, and (c) percent of asperity component of traction for the roller position at $\psi = 0^\circ$.

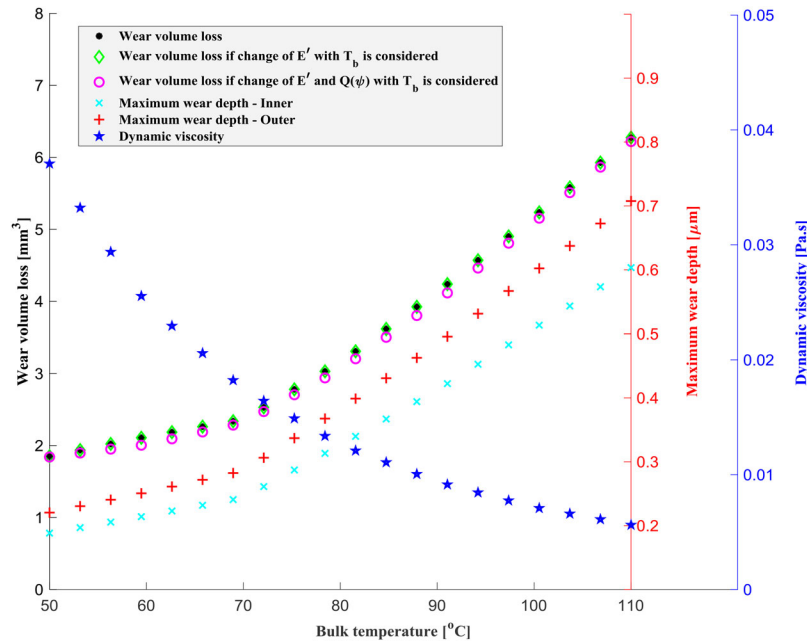


Figure 12. The effect of temperature on wear volume loss and maximum wear depth of a lubricated spherical roller bearing.

contact area, due to elastic modulus change, does not influence the wear significantly so far. For the case of internal load distribution, when the temperature is increased for a given bearing load, a roller is in contact with the races over a slightly larger area for each shaft revolution, which causes a larger

amount of the area of a race to be under wear, but in the central region of the load zone ($\psi = 0^\circ$) the load is reduced, which causes less wear in this area. Thus, the wear volume loss remains almost the same. Therefore, the influence of temperature on bearing wear is mainly due to the reduction of

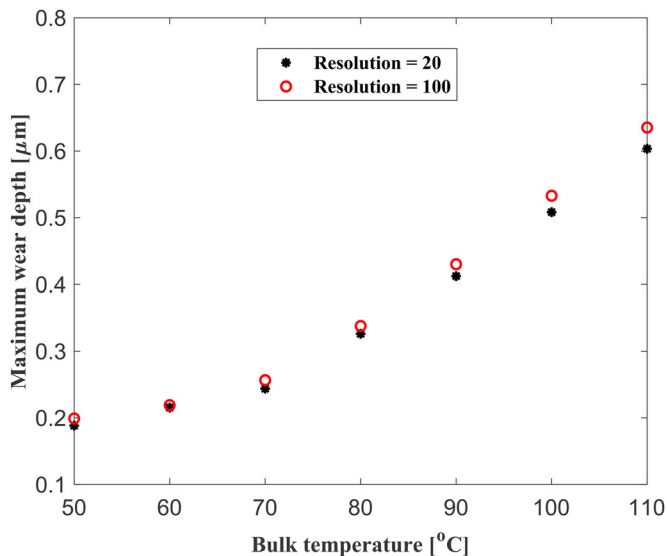


Figure 13. Maximum wear depth of inner race for different resolutions.

film thickness through lubricant viscosity. An increase of wear with temperature was also predicted by Zhao et al. (40) for a Cu-based friction pair lubricated with oil. The reason for wear loss increase is also attributed to the lubricant's rheological properties changing with the temperature. A similar wear rate increases of metallic contact until 300 °C temperature can also be found in reference (66). Other than the lubricant film thickness, the friction and wear coefficients are also influenced by the wear debris (67–70). The impact depends on the properties of the debris particles, such as number, shape and size (68). However, the present study has considered the effect of material removed on friction and wear coefficients. The friction and wear coefficients are interpolated and updated in each iteration based on the coefficients for various usage periods provided in Table 2. The maximum wear depth of each race is also an indicator of how wear changes and Fig. 12 shows a plot of maximum wear depth against temperature. Furthermore, the dynamic viscosity has been replotted in Fig. 12, which shows the wear prediction relation with viscosity at various temperatures. The model's robustness was checked by performing simulations at two resolutions (20 and 100), and Fig. 13 shows the maximum wear depth of the inner race contact for the two different resolutions. The prediction showed that for the higher resolution case (100), the wear depth is slightly higher than for the lower resolution (20). However, the difference in maximum wear depth between the two resolutions is only less than 5%.

Conclusion

This article mainly investigates the wear of the grease lubricated spherical roller bearing for various temperatures. The work also considers the effect of temperature on the internal load distribution, as well as on elastic modulus, and then their subsequent effect on wear. It also investigates the effect of temperature on the load sharing between contact asperity and lubricant, as well as friction/traction behavior of the bearing contact.

For the temperature range considered in this study, the decrease of elastic modulus is not significant (only 3%), which increases the contact patch dimensions by only 1% and decreases the mean contact pressure by approximately 2%. For a given applied radial load to the bearing, the internal load distribution is also changed with temperature. As the bearing temperature is increased from ambient to 110 °C, the load at $\psi = 0^\circ$ is decreased by approximately 4%, and the load zone is increased by approximately 5%. The load distribution change is due to the decrease of bearing operating clearance with the temperature rise. The results show that temperature has a large effect on the friction and wear behavior of the grease-lubricated bearing. When the temperature is increased from 50 to 110 °C, the asperity component of the friction coefficient is increased by almost three times. The percentage change of traction calculation shows that at 50 °C, only approximately 20% of total traction is contributed by the contact asperity; however, this is approximately 80% when the temperature is increased to 110 °C. Prediction shows that wear increases with temperature nonlinearly. For a temperature rise from 50 to 110 °C, the wear volume loss increase is 238%. This is due to the effect of temperature on the film thickness through its effect on viscosity. The wear volume loss changes for the change of internal load distribution and elastic modulus for the temperature range considered in this study are insignificant. The film thickness is lower at a higher temperature, which causes the lubricants to carry a small part of the contact load. The decrease of film thickness promotes the metal-to-metal contact levels, resulting in higher bearing wear. Low operating temperature may be beneficial to achieve a thicker lubricant film and reduce wear, but it increases viscous friction. On the other hand, a high operating temperature reduces viscous friction and lubricant film thickness. Therefore, the bearing should be run at an optimum temperature if both the frictional energy loss and wear are a concern. Finally, it may be concluded that the large prediction of wear volume increases with temperature is mainly related to the lubricant viscosity, with the internal load distribution and elastic modulus having insignificant change. Temperature variation is an unavoidable phenomenon of any dynamic bearing, and this study shows its effect on the grease, elastic modulus, and internal load distribution of a radially loaded bearing and their subsequent effect on wear. This research will help bearing users to optimize the operational condition. It can also be a benchmark for other researchers to investigate the temperature effect on bearing failure mechanisms from a broader perspective.

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